

Assessment – 3
CSA1717 – Artificial Intelligence

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QUESTION 1:

Graph & Heuristic

Node h(n)	
A	7
B	6
C	4
D	3
E	2
G	0

Edges

- $A \rightarrow B$ (2), $A \rightarrow C$ (4)
- $B \rightarrow C$ (3), $B \rightarrow D$ (7), $B \rightarrow E$ (2)
- $C \rightarrow E$ (3)
- $D \rightarrow E$ (2)
- $E \rightarrow G$ (2)

A^* Dry Run from $A \rightarrow G^*$

Iteration 1

Current Node: A

Open List: A

Closed List: —

Node g(n) h(n) f(n)

A 0 7 7

Expand A → neighbors

- B: $g=2, h=6 \rightarrow f=8$
- C: $g=4, h=4 \rightarrow f=8$

Open List: B(8), C(8)

Closed List: A

Iteration 2

Current Node: B (tie broken alphabetically)

Open List: C

Closed List: A, B

Node g(n) h(n) f(n)

B 2 6 8

Expand B → neighbors

- C: new $g=2+3=5 \rightarrow f=5+4=9$ (existing $f=8 \rightarrow$ keep old)
- D: $g=2+7=9 \rightarrow f=9+3=12$
- E: $g=2+2=4 \rightarrow f=4+2=6$

Open List: C(8), E(6), D(12)

Sorted: E(6), C(8), D(12)

Iteration 3

Current Node: E

Open List: C, D

Closed List: A, B, E

Node g(n) h(n) f(n)

E 4 2 6

Expand E $\rightarrow$ neighbors

- G: $g=4+2=6 \rightarrow h=0 \rightarrow f=6$

Open List: G(6), C(8), D(12)

Iteration 4

Current Node: G

Goal Reached

Optimal Path Found

A $\rightarrow$ B $\rightarrow$ E $\rightarrow$ G

Total Path Cost

- A $\rightarrow$ B = 2
- B $\rightarrow$ E = 2
- E $\rightarrow$ G = 2

Total Cost = 2 + 2 + 2 = 6

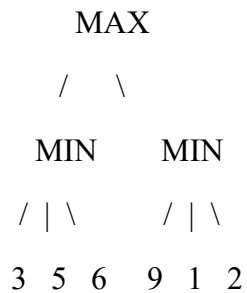
FINAL ANSWER – QUESTION 1

Optimal Path: A $\rightarrow$ B $\rightarrow$ E $\rightarrow$ G

Total Cost: 6

QUESTION 2:

Game Tree



Dry Run Table:

Step 1 – MAX at root

$$\alpha = -\infty \quad \beta = +\infty$$

Explore Left MIN subtree first

Step 2 – MIN (Left Subtree)

$$\alpha = -\infty \quad \beta = +\infty$$

Children: 3, 5, 6

- Evaluate 3
 - MIN chooses min $\rightarrow 3$
 - $\beta = 3$
- Evaluate 5
 - MIN compares: $\min(3, 5) = 3$
 - β stays 3
- Evaluate 6
 - MIN compares: $\min(3, 6) = 3$
 - β stays 3

Left MIN subtree value = 3

MAX updates: $\alpha = \max(-\infty, 3) = 3$

Step 3 – MIN (Right Subtree)

$\alpha = 3$ $\beta = +\infty$

Children: 9, 1, 2

- Evaluate 9
 - MIN picks 9
 - $\beta = 9$

Check pruning condition:

If $\beta \leq \alpha \rightarrow$ prune $9 \leq 3 \rightarrow \text{FALSE} \rightarrow$ continue

- Evaluate 1
 - MIN picks $\min(9, 1) = 1$
 - $\beta = 1$

Check pruning:

$1 \leq 3 \rightarrow \text{TRUE} \rightarrow$ prune remaining nodes

Node 2 is PRUNED

Right MIN subtree value = 1

Step 4 – MAX chooses best

MAX compares: $\max(3, 1) = 3$

FINAL ANSWERS:

Best Move for MAX: Left Subtree

Final Minimax Value: 3

Pruned Nodes: 2
